

Show all of your work in the space provided. Full credit will not be awarded for correct answers unless sufficient supporting work is presented.

1. (2 pts) Give a quick definition of what it means to say a vector space, V , is finite dimensional.

Any basis for V consists of a finite number of vectors.

2. (12 pts) Let A be a 12×10 matrix with three **non-pivot** columns. What are the following values?

a) Rank $A = 7$

b) Nullity $A = 3$

c) Dimension of $\text{Col}(A) = 7$

d) Dimension of $\text{Col}(A^T) = 7$

e) Dimension of $\text{Row}(A) = 7$

f) Dimension of $\text{Nul}(A^T) = 5$

3. (10 pts) Let $A = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$.

a) What is the characteristic equation for A ?

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 4 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 4 = 4 - 4\lambda + \lambda^2 - 4 = \lambda^2 - 4\lambda$$

$$\boxed{\lambda^2 - 4\lambda = 0}$$

b) What are the eigenvalues for A ?

$$\lambda(\lambda - 4) = 0 \quad \text{so} \quad \boxed{\lambda = 0 \quad \text{or} \quad \lambda = 4}$$

4. (10 pts) A scientist finds a solution to a nonhomogeneous system $Ax = b$ of 10 equations with 8 unknowns. Using this fact, **decide if each statement below is impossible, possible but not guaranteed, or must be true.** Circle the correct choice and give a very brief explanation of your answer.

a) The system has a solution for all choices of b .

impossible

possible but not guaranteed

must be true

Explain: There are at most 8 pivots so you cannot have a pivot in every row. This means there will definitely be some vectors b for which the system is inconsistent.

b) For this particular choice of b the system has infinitely many solutions.

impossible

possible but not guaranteed

must be true

Explain: If there are 8 pivots we have a pivot in every column so this system would have a unique solution. But there could be free variables so the system could have an infinite solution set.

5. (16 pts) Let $A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{bmatrix}$. The eigenvalues of A are 1 and 5.

Is A diagonalizable? If so, find a matrix P such that $A = PDP^{-1}$.

$$\lambda = 1 \quad A - I = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & -1 \\ -1 & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = -2x_2 + x_3 \\ x_2 = x_2 \\ x_3 = x_3 \end{array}$$

Basis for eigenspace for $\lambda = 1$ is $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$\lambda = 2 \quad A - 5I = \begin{bmatrix} -3 & 2 & -1 \\ 1 & -2 & -1 \\ -1 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = -x_3 \\ x_2 = -x_3 \\ x_3 = x_3 \end{array}$$

Basis for eigenspace for $\lambda = 2$ is $\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$

A is diagonalizable

$$P = \begin{bmatrix} -2 & 1 & -1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

6. (12 pts) Let A be a 3×3 matrix with eigenvalues 1, -2, 3, with corresponding

eigenvectors $v_1 = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$, and $v_3 = \begin{bmatrix} -3 \\ -3 \\ 0 \end{bmatrix}$. Let $x = \begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix}$

a) Find the coordinate vector for x relative to the basis $B = \{v_1, v_2, v_3\}$.

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 8 \\ 0 & 1 & -3 & 5 \\ -3 & 5 & 0 & 7 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

$$\begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix}_B = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

b) Find the value of A^3x .

$$\begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} - \begin{bmatrix} -3 \\ -3 \\ 0 \end{bmatrix}$$

$$A^3 \begin{bmatrix} 8 \\ 5 \\ 7 \end{bmatrix} = A^3 \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + 2A^3 \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} - A^3 \begin{bmatrix} -3 \\ -3 \\ 0 \end{bmatrix}$$

$$= 1^3 \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + 2 \cdot (-2)^3 \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} - 3^3 \begin{bmatrix} -3 \\ -3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + \begin{bmatrix} -32 \\ -16 \\ -80 \end{bmatrix} + \begin{bmatrix} 81 \\ 81 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 50 \\ 65 \\ -83 \end{bmatrix}$$

7. (8 pts) If x is an eigenvector for A with eigenvalue λ , show that x is also an eigenvector for $5A$.

$$\text{Given } A\vec{x} = \lambda\vec{x} \quad \text{then } (5A)\vec{x} = 5(A\vec{x}) = 5(\lambda\vec{x}) = (5\lambda)\vec{x}$$

So $\vec{x}$ is an eigenvector with eigenvalue 5λ

8. (4 pts) Let W be a subspace of $\mathbb{R}^n$. What does it mean for the vector x to be in the orthogonal complement of W ?

The dot product of $\vec{x}$ with any vector in W will be 0.

$\vec{x}$ is orthogonal to every vector in W .

9. (10 pts) Let $\mathbf{u} = \begin{bmatrix} 6 \\ 4 \\ -3 \end{bmatrix}$.

a) Give an example of a nonzero vector that is orthogonal to $\mathbf{u}$.

many answers are correct $\begin{bmatrix} -4 \\ 6 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix}$
Any vector $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ such that $6a + 4b - 3c = 0$

b) Find a unit vector in the opposite direction of $\mathbf{u}$.

$$\frac{-\mathbf{u}}{\|\mathbf{u}\|} = \frac{1}{\sqrt{36+16+9}} \begin{bmatrix} -6 \\ -4 \\ 3 \end{bmatrix} = \frac{1}{\sqrt{61}} \begin{bmatrix} -6 \\ -4 \\ 3 \end{bmatrix} \quad \text{or} \quad \frac{-1}{\sqrt{61}} \begin{bmatrix} 6 \\ 4 \\ -3 \end{bmatrix}$$

10. Let W be the subspace spanned by the orthogonal vectors $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$

a) (6 pts) What is the projection of the vector $y = \begin{bmatrix} 3 \\ -2 \\ 4 \\ -3 \end{bmatrix}$ onto W ?

$$\hat{y} = \frac{y \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{y \cdot v_2}{v_2 \cdot v_2} v_2 = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \frac{12}{4} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 7/2 \\ -5/2 \\ 3 \\ -3 \end{bmatrix} \text{ or } \begin{bmatrix} 3.5 \\ -2.5 \\ 3 \\ -3 \end{bmatrix}$$

b) (4 pts) Write y as the sum of a vector in W and a vector in $W^\perp$.

$$y = \hat{y} + y - \hat{y} = \begin{bmatrix} 7/2 \\ -5/2 \\ 3 \\ -3 \end{bmatrix} + \begin{bmatrix} -1/2 \\ 1/2 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{or} \quad \begin{bmatrix} 3.5 \\ -2.5 \\ 3 \\ -3 \end{bmatrix} + \begin{bmatrix} -.5 \\ .5 \\ 1 \\ 0 \end{bmatrix}$$

11. (6 pts) Let A be any $m \times n$ matrix. The four fundamental subspaces associated with matrix A are listed in the column on the left. Several other subspaces associated with A are listed in the column on the right. Draw a line between subspaces that are equal.

